```
Feature Description:
                 age           sex           bmi            bp            s1  \
count  4.420000e+02  4.420000e+02  4.420000e+02  4.420000e+02  4.420000e+02
mean  -2.511817e-19  1.230790e-17 -2.245564e-16 -4.797570e-17 -1.381499e-17
std    4.761905e-02  4.761905e-02  4.761905e-02  4.761905e-02  4.761905e-02
min   -1.072256e-01 -4.464164e-02 -9.027530e-02 -1.123988e-01 -1.267807e-01
25%   -3.729927e-02 -4.464164e-02 -3.422907e-02 -3.665608e-02 -3.424784e-02
50%    5.383060e-03 -4.464164e-02 -7.283766e-03 -5.670422e-03 -4.320866e-03
75%    3.807591e-02  5.068012e-02  3.124802e-02  3.564379e-02  2.835801e-02
max    1.107267e-01  5.068012e-02  1.705552e-01  1.320436e-01  1.539137e-01

                 s2            s3            s4            s5            s6
count  4.420000e+02  4.420000e+02  4.420000e+02  4.420000e+02  4.420000e+02
mean   3.918434e-17 -5.777179e-18 -9.042540e-18  9.293722e-17  1.130318e-17
std    4.761905e-02  4.761905e-02  4.761905e-02  4.761905e-02  4.761905e-02
min   -1.156131e-01 -1.023071e-01 -7.639450e-02 -1.260971e-01 -1.377672e-01
25%   -3.035840e-02 -3.511716e-02 -3.949338e-02 -3.324559e-02 -3.317903e-02
50%   -3.819065e-03 -6.584468e-03 -2.592262e-03 -1.947171e-03 -1.077698e-03
75%    2.984439e-02  2.931150e-02  3.430886e-02  3.243232e-02  2.791705e-02
max    1.987880e-01  1.811791e-01  1.852344e-01  1.335973e-01  1.356118e-01

Target Description:
 count    442.000000
mean     152.133484
std       77.093005
min       25.000000
25%       87.000000
50%      140.500000
75%      211.500000
max      346.000000
Name: progression, dtype: float64

Feature Names: ['age', 'sex', 'bmi', 'bp', 's1', 's2', 's3', 's4', 's5', 's6']

Training set size: 353
Testing set size: 89
```

스크린샷

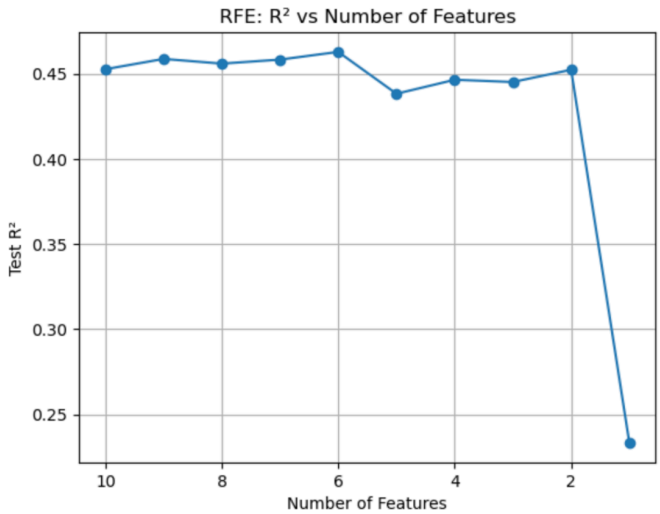
Task 1

This dataset consists of a total of 442 samples and uses 10 input variables, including age, sex, BMI, bp, and six serum measurements (s1–s6). The target variable of this dataset is a continuous value representing the progression of the disease one year after the baseline. From this, we can understand that a higher value indicates a more severe progression of diabetes. The dataset was divided into 80% training data and 20% testing data.

```
R^2 Score: 0.45260276297191937
```

Task 2

According to the analysis results, the model's prediction accuracy was approximately 0.45, indicating that it explains the progression of diabetes to some extent. From this model, we can see that BMI and a specific serum measurement (s5) have the greatest influence on the outcome. Therefore, it can be predicted that these two factors—obesity level and certain blood measurement data—may pose a higher risk for the progression of diabetes compared to other factors.




```
Optimal number of features: 10
Selected features: [np.str_('age'), np.str_('sex'), np.str_('bmi'), np.str_('bp'), np.str_('s1'), np.str_('s2'), np.str_('s3'), np.str_
('s4'), np.str_('s5'), np.str_('s6')]
```

Task 3

When using between 2 and 10 features, the prediction score remained similar at around 0.45, but it was observed that the best performance was achieved when all 10 features were used. On the other hand, when only one feature was used, the performance dropped significantly, and there was no performance improvement above the threshold (0.01). Therefore, it was concluded that using all 10 features is the most effective approach for predicting the progression of diabetes.

	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6
n_features										
10	37.904	-241.9644	542.4288	347.7038	-931.4888	518.0623	163.4200	275.3179	736.1989	48.6707
9	NaN	-236.6496	542.7995	354.2114	-936.3506	528.7966	167.8004	270.3965	744.4474	53.3505
8	NaN	-233.7547	550.7444	363.7918	-947.8231	541.5858	172.2506	277.7411	761.9212	NaN
7	NaN	-235.3642	551.8664	362.3561	-660.6432	343.3481	NaN	185.1408	664.7746	NaN
6	NaN	-215.2674	557.3142	350.1787	-851.5157	591.0933	NaN	NaN	803.1213	NaN
5	NaN	NaN	597.8927	306.6479	-655.5606	409.6222	NaN	NaN	728.6436	NaN
4	NaN	NaN	691.4601	NaN	-592.9779	362.9503	NaN	NaN	783.1685	NaN
3	NaN	NaN	737.6856	NaN	-228.3399	NaN	NaN	NaN	680.2247	NaN
2	NaN	NaN	732.1090	NaN	NaN	NaN	NaN	NaN	562.2265	NaN
1	NaN	NaN	998.5777	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Task 4

Based on the results above, we were able to examine the importance of the variables. The three most important factors affecting diabetes progression were BMI, s5, and blood pressure. In particular, higher BMI was associated with worse diabetes progression, and s5 and blood pressure also played important roles in prediction. Although using all 10 features gave slightly higher accuracy, focusing on these key variables maintained similar performance while making it much easier to understand what factors contribute to diabetes progression.

Task 5

1. What did you learn about feature selection using RFE?

By removing unnecessary information one by one, I learned that keeping only the important features can maintain the same prediction performance. As the model became simpler, it was much easier to understand why the results turned out that way.

2. How does RFE compare to other feature selection methods like LASSO in terms of methodology and results?

RFE works like a “survival” method where low-performing features are eliminated one by one, so it is easy to observe the process but takes more time. In contrast, LASSO works like a “penalty” method that reduces unimportant features to zero at once, making it much faster and more efficient.

3. What insights can you draw about the dataset from the selected features?

In the end, BMI, s5, and blood pressure played the most important roles in diabetes progression. Rather than many complex variables, these three indicators serve as the key factors in predicting diabetes risk.